Predicting 30-Day Hospital Readmission for Diabetic Patients

Phase 1: Exploratory Data Analysis, Preprocessing & Feature Engineering

Machine Learning Course — Spring 2026

Presentation Outline

1. Problem Definition & Motivation
2. Dataset Overview
3. Exploratory Data Analysis (EDA)
4. Preprocessing Pipeline
5. Feature Engineering
6. Train-Test Split & Scaling
7. Summary & Phase 2 Preview

Key Takeaway

We will move from **raw data** to **model-ready features** in 30 slides.

Hospital Readmission: A $26 Billion Problem

- 30-day hospital readmissions cost the US healthcare system **~$26 billion annually**.
- The Hospital Readmissions Reduction Program (HRRP) **penalizes** hospitals with excess readmission rates.
- **Diabetic patients** are especially vulnerable due to complex medication regimens, multiple comorbidities, and glycemic management challenges.
- Early identification of high-risk patients at discharge enables **targeted interventions**.

Annual System Cost

$26 Billion

Admit > Discharge > < 30 Days > **Readmit**

Key Takeaway

Reducing avoidable readmissions has large clinical and financial impact.

Why ML for Readmission Prediction?

Traditional Approach

- ✕ Rule-based systems (e.g., LACE index) use only **3-5 variables**
- ✕ Limited ability to capture **complex interactions**
- ✕ **Static thresholds**, with no dynamic adaptation to new data

ML Approach

- ✓ Integrates **50+ features** simultaneously
- ✓ Captures **non-linear relationships** and feature interactions
- ✓ **Data-driven** and easily adaptable to new patient populations

Our approach: Build a binary classifier using clinical encounter data from 130 US hospitals.

Key Takeaway

Machine Learning can synthesize many weak signals into highly accurate and actionable risk scores.

Dataset Overview — Diabetes 130-US Hospitals

101,766
Encounters × 50 features

130 Hospitals
Across US (1999-2008)

Target
Readmitted < 30 days

UCI Repo
Strack et al., 2014

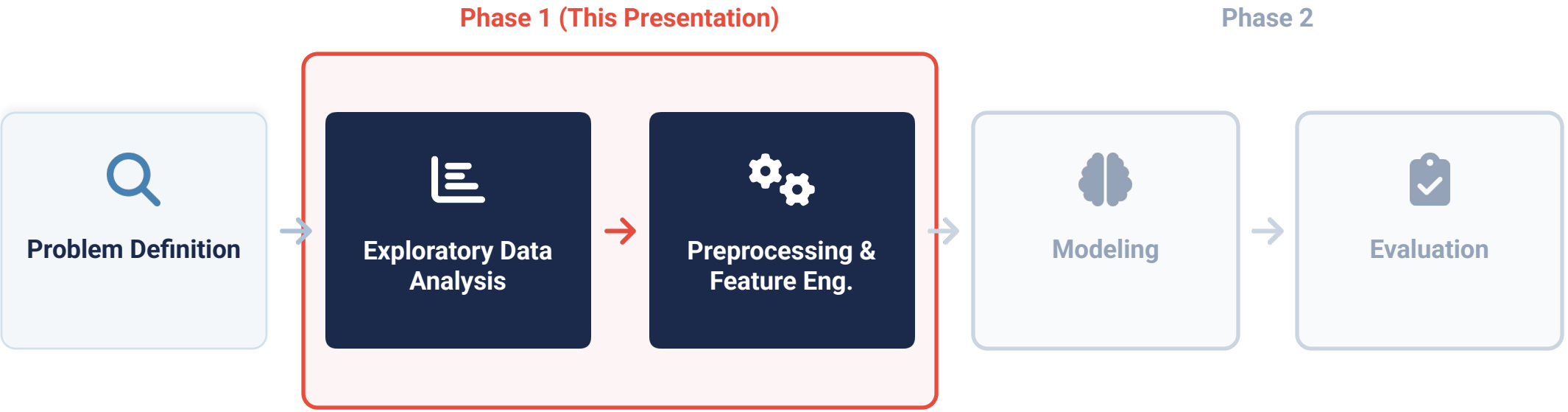
Feature Groups Overview

Group	Features	Scale
Demographics	race, gender, age	Nominal, Ordinal
Admission	admission_type, discharge_disposition, admission_source	Nominal
Encounter	time_in_hospital, num_lab_procedures, num_medications, etc.	Ratio
Diagnoses	diag_1, diag_2, diag_3 (ICD-9 codes)	Nominal
Lab Results	max_glu_serum, A1Cresult	Ordinal
Medications	23 specific diabetes drug columns	Ordinal
Flags	change, diabetesMed	Binary

Key Takeaway

A rich, multi-domain EHR dataset enables comprehensive modeling of patient risk factors across demographics, utilization, and treatment.

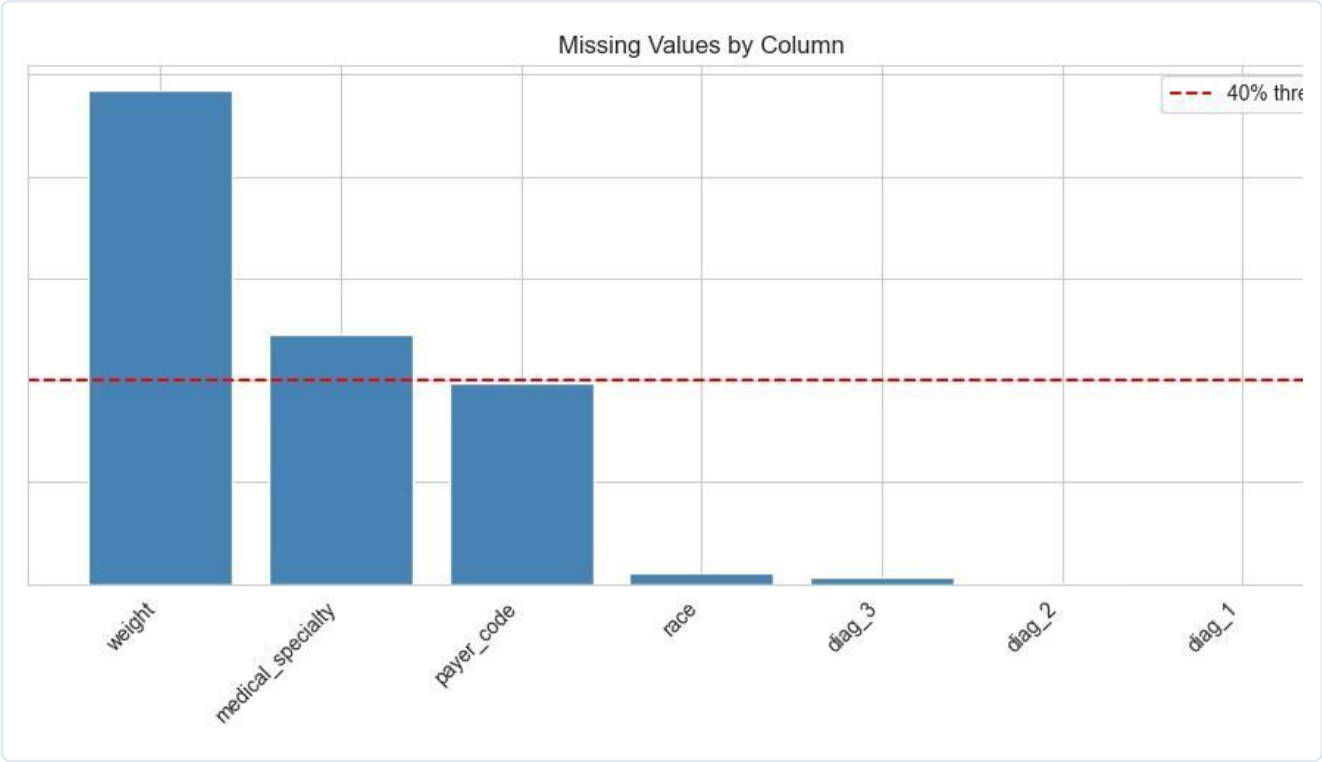
The Machine Learning Pipeline



Key Takeaway

A clear, well-defined pipeline prevents data leakage and structures systematic experimentation across both phases of the project.

Missing Values: Extent & Treatment



Missingness Profile & Strategy

Exceeds Threshold (> 40%)

ACTION: DROP

weight: 97% missing

medical_specialty: 49% missing

payer_code: 40% missing

Below Threshold (< 40%)

ACTION: IMPUTE

race: ~2% missing (mode imputation)

diag_1, diag_2, diag_3: < 2% missing (assign 'Missing' category)

Key Takeaway

Columns exceeding 40% missingness are dropped. Remaining values are imputed with the mode or a 'Missing' category to preserve valuable data while reducing noise.

Rubin's Missing Data Taxonomy Applied

Column	Missing %	Type	Reasoning	Action
weight	97%	MNAR	Recording depends on clinical workflow and patient factors	Drop
medical_specialty	49%	MAR	Missingness correlates with admission pathway	Drop
payer_code	40%	MAR	Administrative, not clinical	Drop
race	2%	MCAR	Random omissions	Impute (mode)
diag_1/2/3	<2%	MAR	Simpler cases have fewer secondary diagnoses	Impute ('Missing')

MCAR: Missing Completely At Random **MAR:** Missing At Random **MNAR:** Missing Not At Random

Key Takeaway

Tailored actions based on the missing data mechanism preserve dataset validity and avoid systemic bias.

Preventing Data Leakage via Patient Deduplication

The Problem

- Dataset contains **101,766 encounters** but only **~71,500 unique patients**.
- The same patient could randomly appear in both the **train AND test sets**.
- The model would **memorize patient-specific patterns** instead of learning general risk factors.

The Solution

- Keep only the **first encounter** per patient based on encounter IDs.
- Reduces dataset by ~30%, but ensures **total independence**.
- Preserves the **validity of out-of-sample evaluation** metrics.

Key Takeaway

This is the first of three critical data leakage safeguards (along with train-only scaler fitting and stratified splitting).

Target Variable — Class Imbalance

The Accuracy Paradox

A model that always predicts 'not readmitted' achieves **91% accuracy** but catches ZERO at-risk patients.

Evaluation Metrics Strategy

Accuracy: Misleading for imbalanced data

Precision: Of flagged patients, how many are truly at risk?

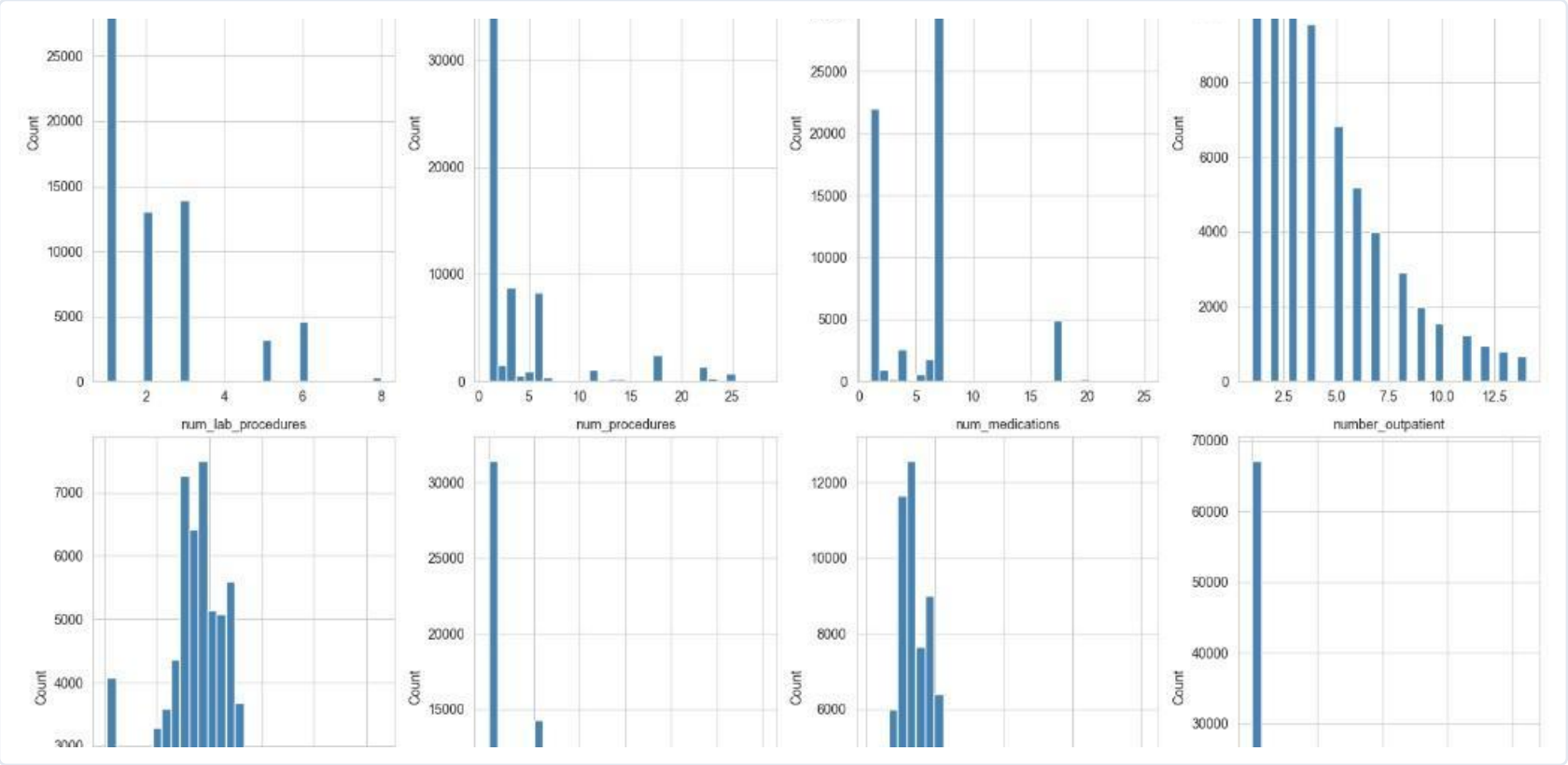
Recall: Of at-risk patients, how many did we catch?

F1-Score: Harmonic mean balancing precision and recall

Key Takeaway

Model must handle severe imbalance (91% vs 9%) to be clinically useful, requiring specialized metrics like Precision, Recall, and F1.

Univariate Analysis — Numeric Features



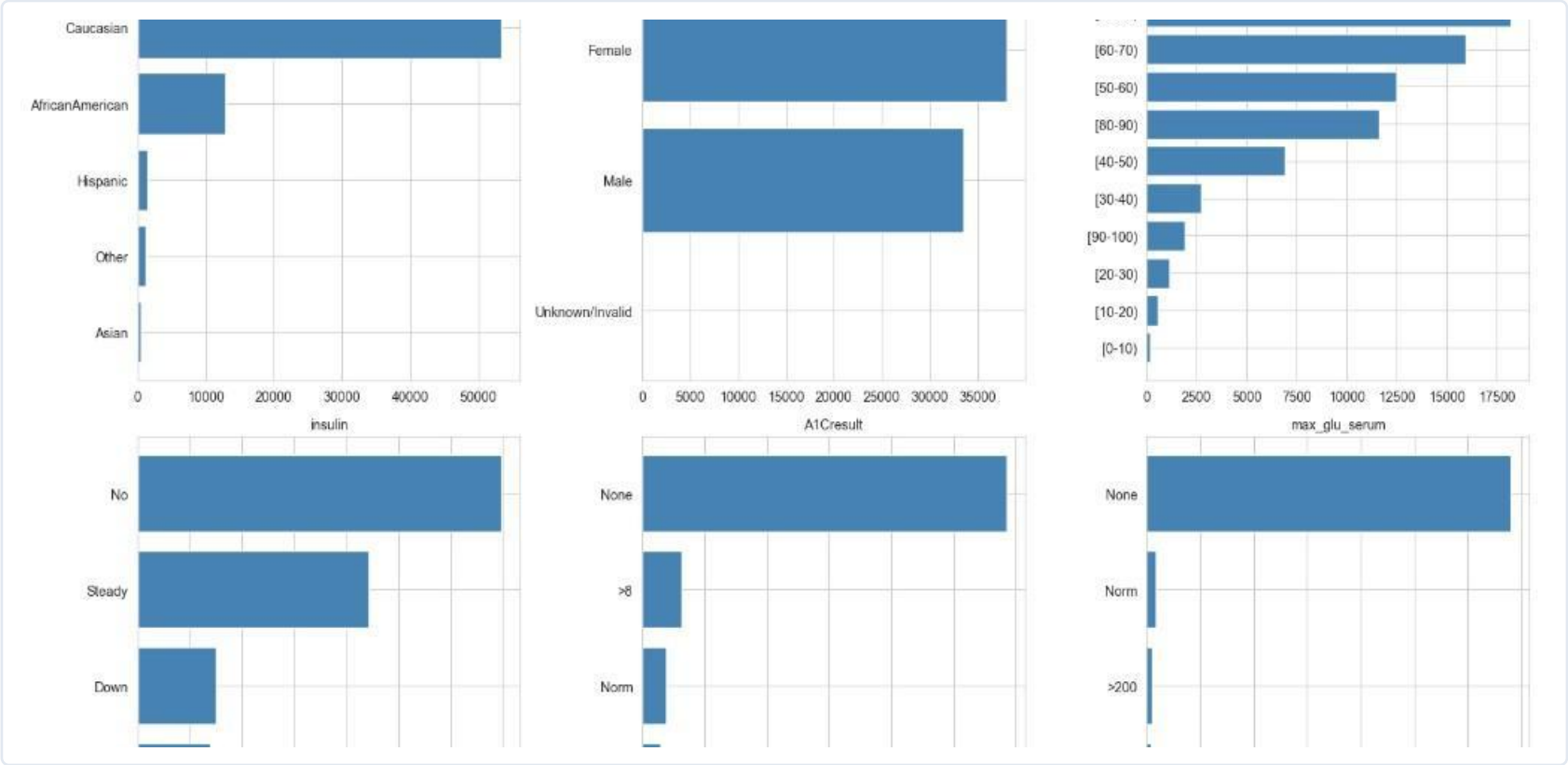
Key Observations

- Most distributions are right-skewed, requiring transformation.
- Utilization features (outpatient, emergency, inpatient) are zero-inflated with extreme tails.

Key Takeaway

Skew and zero inflation inform transformations and robust modeling choices to manage long tails.

Univariate Analysis — Categorical Features



Key Observations

- Race is dominated by Caucasian; age peaks at 70–80.
- A1C and glucose serum tests were mostly 'None'.

Key Takeaway

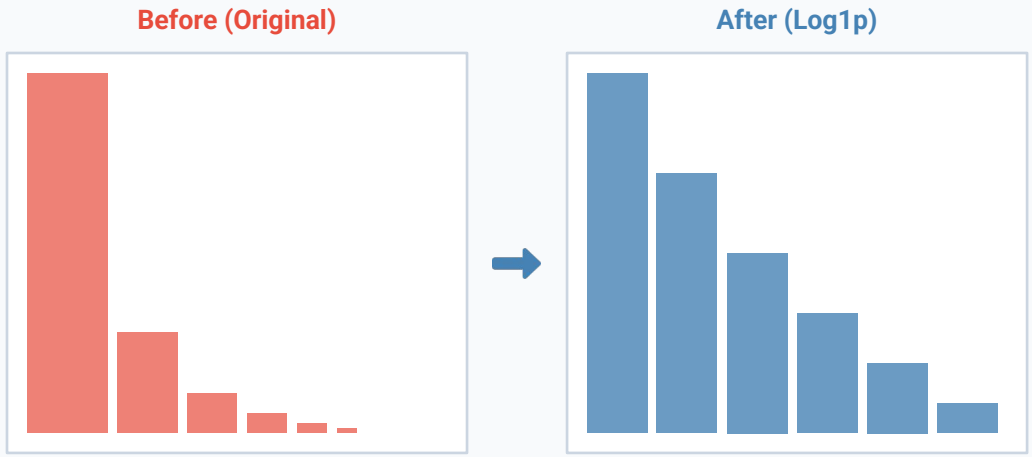
'Test not performed' ('None') categories are informative in themselves and highlight testing patterns.

Feature Skewness & Transformation Candidates

Top Skewed Features

Feature	Skewness	Action
number_outpatient	>5	Log1p transform
number_emergency	>5	Log1p transform
number_inpatient	>3	Log1p transform
num_procedures	~1.5	Monitor

Transformation Effect (number_inpatient)

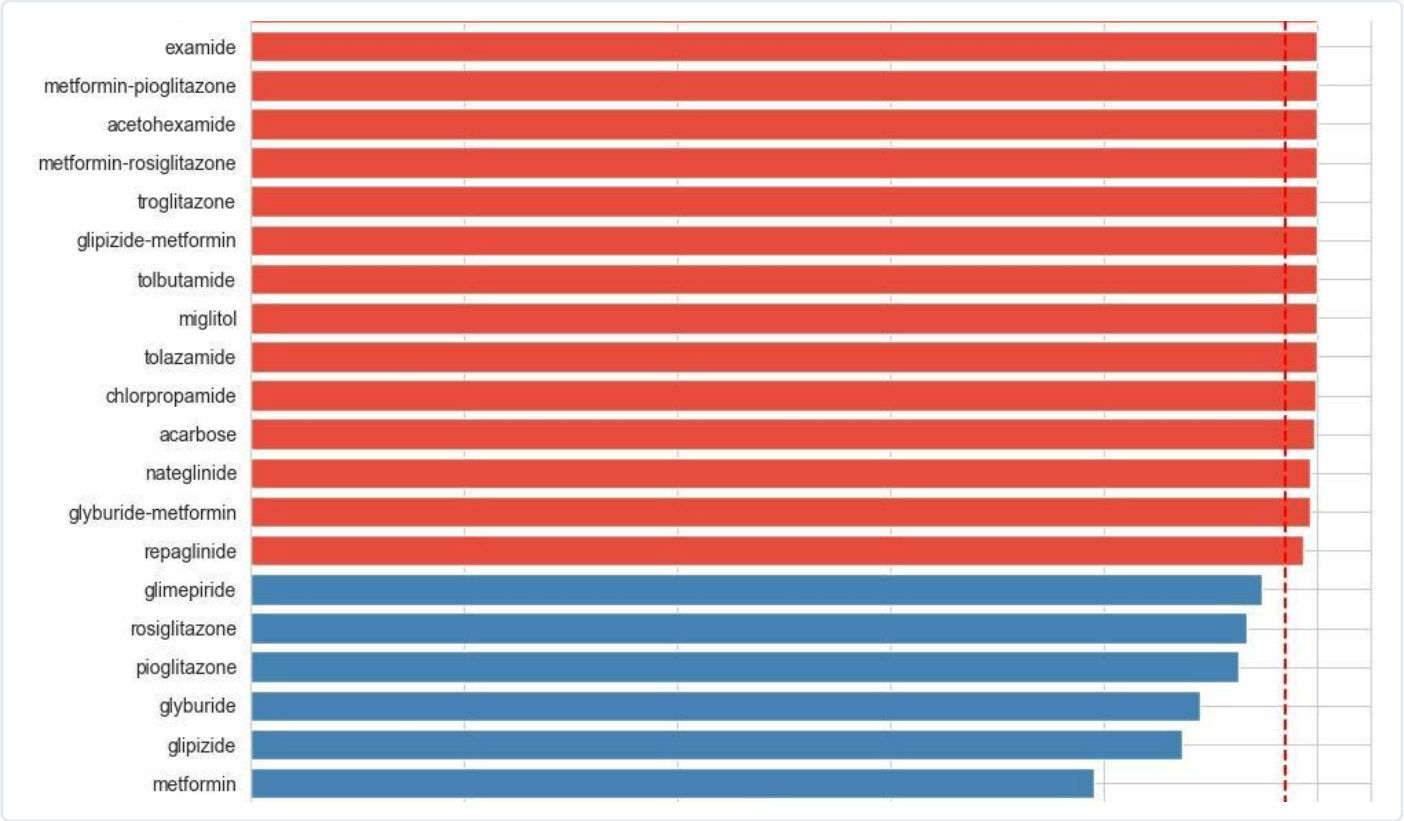


Heavily skewed features distort distance-based algorithms. Log1p compresses the right tail while preserving zero values.

Key Takeaway

Log1p transformation reduces the leverage of extreme values in highly skewed features while keeping zero counts intact.

16 of 23 Medication Columns Are Uninformative



Near-Zero Variance (NZV)

Extreme Imbalance Examples

citoglipton: **100%** No
examide: **100%** No
troglitazone: **99.8%** No
acetohexamide: **99.7%** No

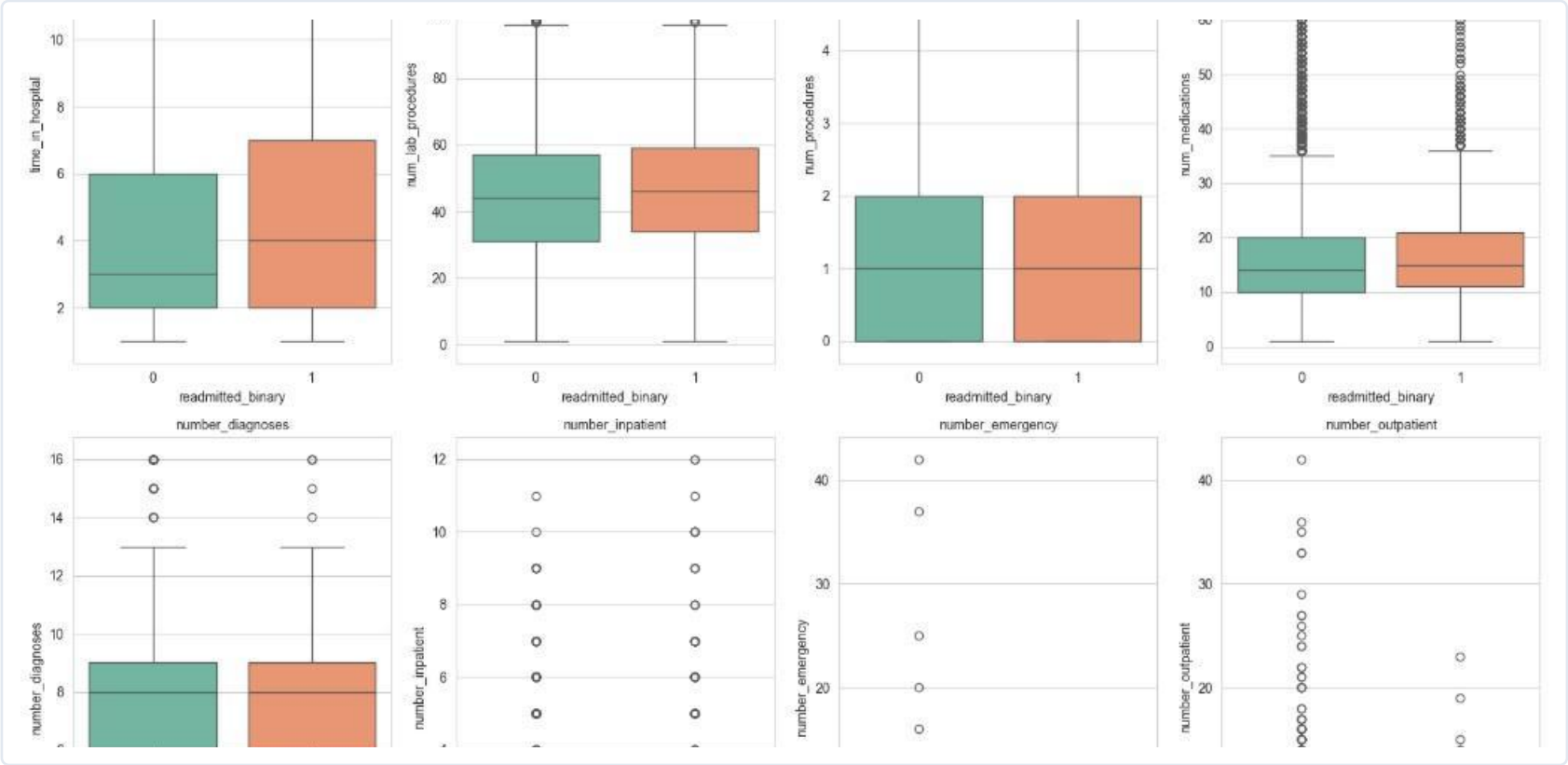
Action Taken

Features with >97% identical values have near-zero entropy and carry negligible predictive information. They are dropped to reduce dimensionality.

Key Takeaway

Removing low-entropy features simplifies the model without losing signal.

Bivariate Analysis — Boxplots by Readmission



Key Differences

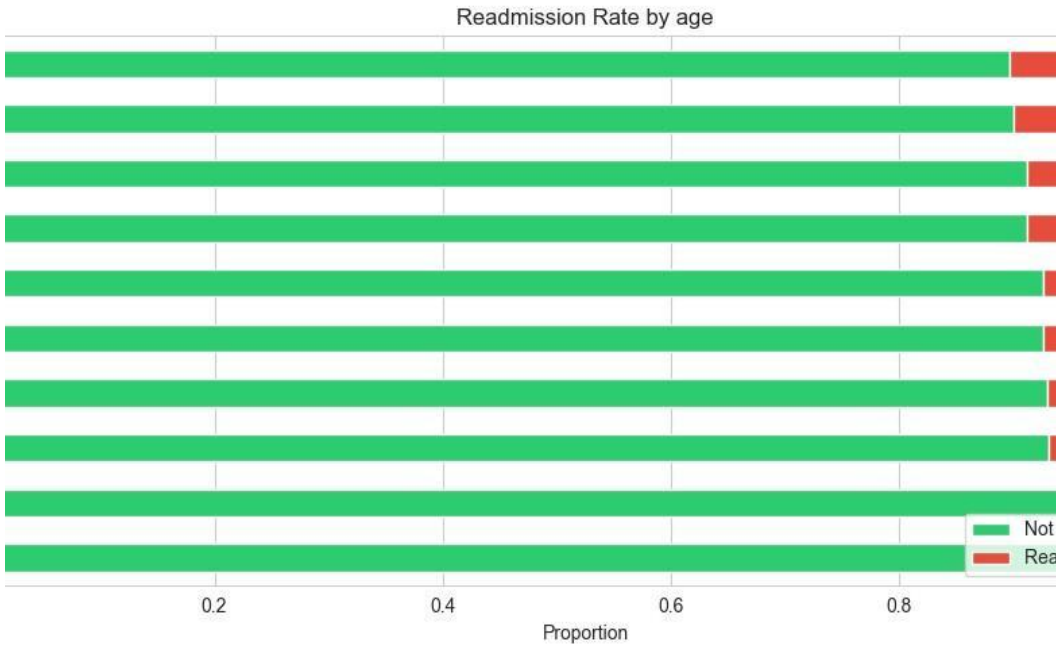
- **Readmitted patients** have higher prior inpatient visits.
- More prior ER visits are observed for the readmitted group.
- Slightly longer hospital stays on average.

Key Takeaway

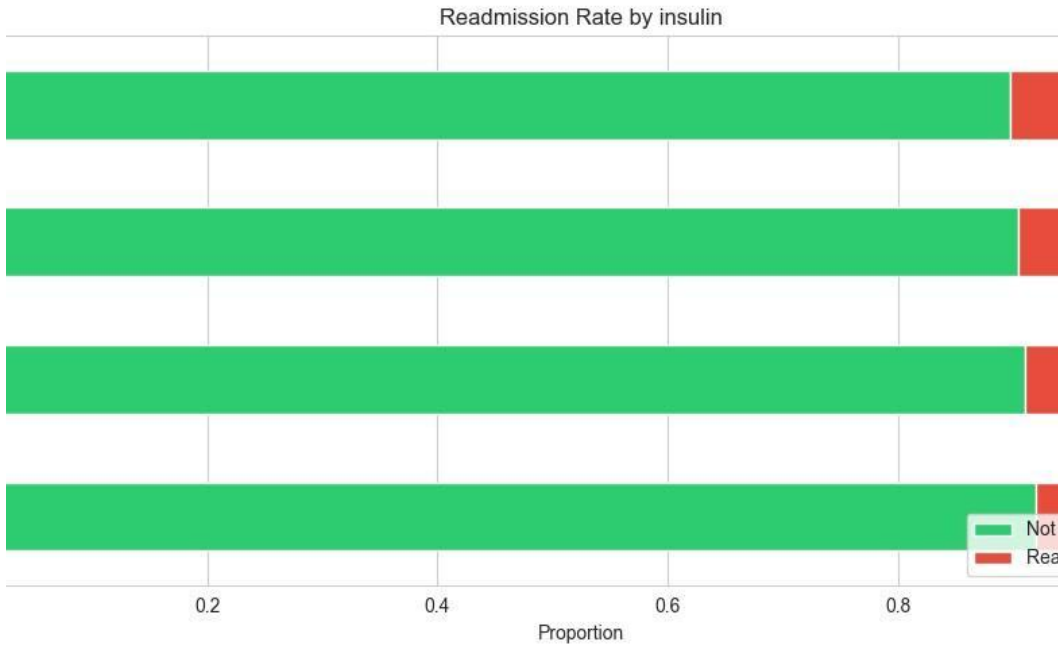
Prior utilization history (inpatient and ER visits) is associated with higher readmission risk, making it a critical signal.

Readmission Rate Across Subgroups

Readmission Rate by Age



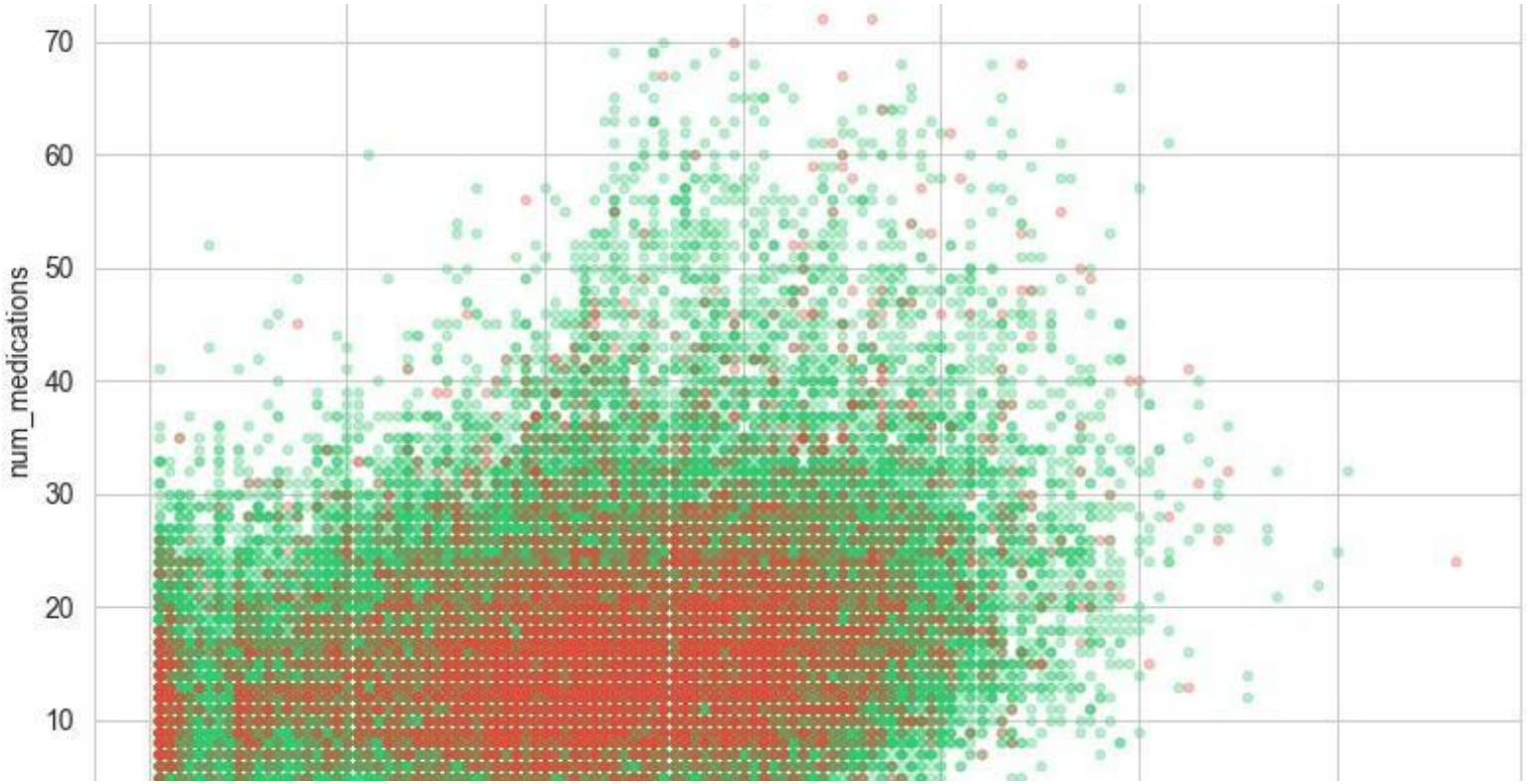
Readmission Rate by Insulin Status



Key Takeaway

Readmission rates vary across age groups and insulin management patterns. 'Down' or 'Up' insulin status indicates higher risk than 'No' insulin, providing strong demographic and clinical signals.

No Simple Linear Boundary Separates the Classes



Class Overlap Analysis

Observation

The scatter plot demonstrates heavy overlap between the readmitted (red) and not readmitted (green) classes across two key numeric features.

Implication 1: No single pair of features cleanly separates the high-risk patients.

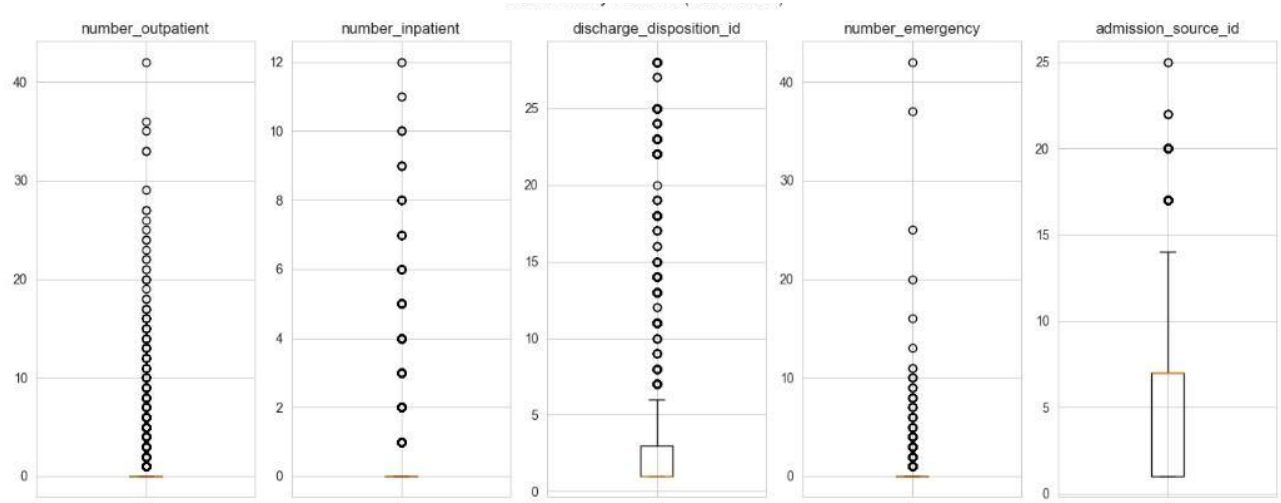
Implication 2: Non-linear models and complex feature interactions will be essential for predictive performance.

Key Takeaway

Linear separation is impossible; we anticipate non-linear algorithms like Decision Trees to be crucial in Phase 2.

Clinical vs Statistical Outliers

Statistical Outliers (IQR Method)



% Flagged as Statistical Outliers:

- number_outpatient: **15%**
- number_emergency: **12%**
- number_inpatient: **8%**

Clinical Interpretation

- These are **NOT** data errors or invalid entries.
- They represent genuine high-risk **"frequent utilizer"** patients with complex medical needs.
- Removing them would artificially bias the model against detecting the most vulnerable, high-risk individuals.

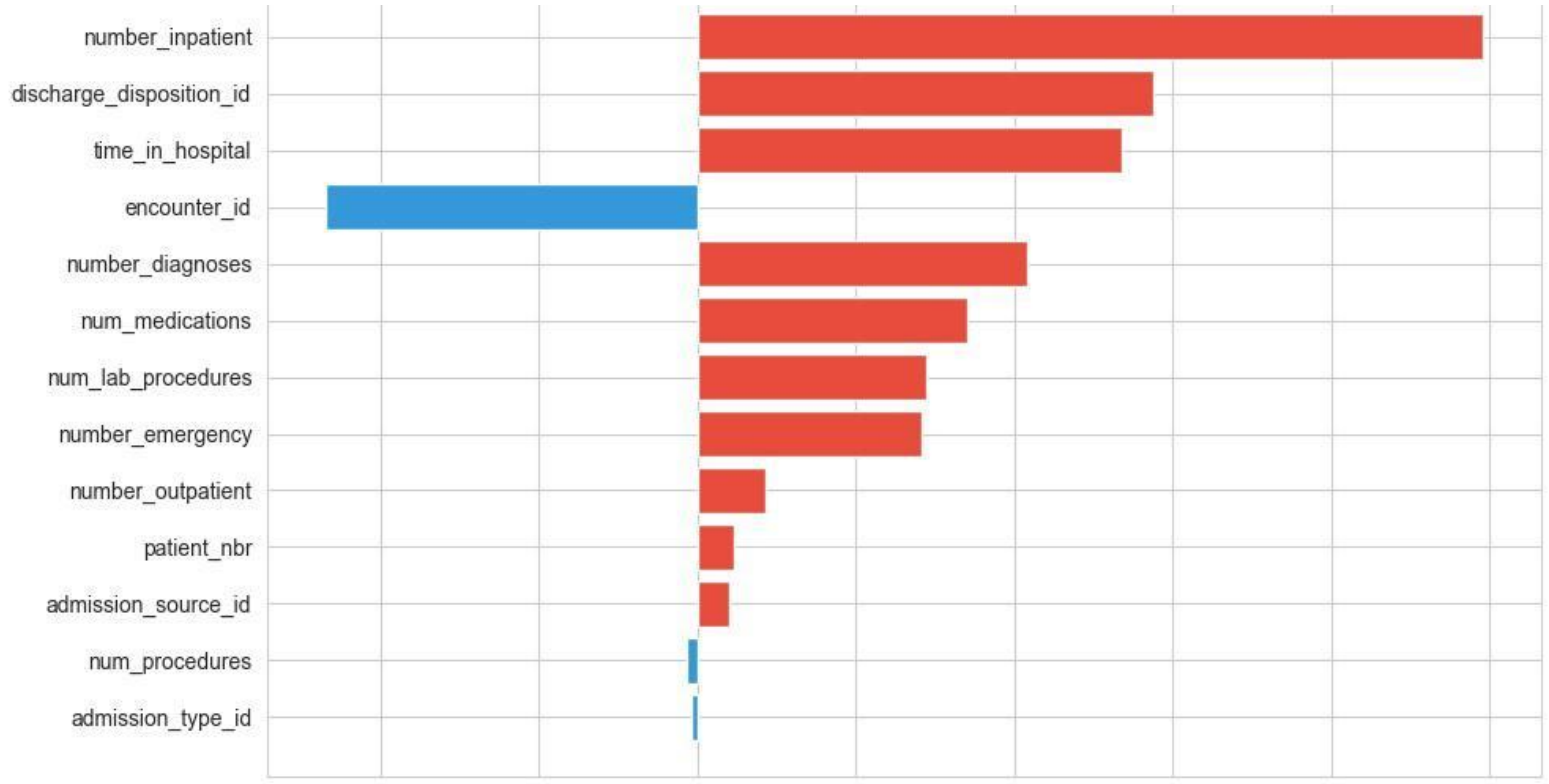
Treatment Strategy

Winsorize (cap at 1st/99th percentile): preserves signal while limiting extreme leverage.

Key Takeaway

Preserve clinically meaningful extremes while limiting statistical leverage by Winsorizing (capping) instead of dropping.

Feature Correlations with Readmission



Correlation Insights

Individual Pearson correlations are weak (|r| < 0.1). Power comes from combining features.

Pearson only measures linear relationships. Non-linear associations may be stronger.

Full heatmap analysis (not shown) guided multicollinearity pruning.

Key Takeaway

Predictive power will come from combining many weak signals, not relying on a single dominant feature.

Seven Key Findings from EDA

1. Severe Class Imbalance

~9% positive class requires using precision, recall, and F1-score rather than simple accuracy.

2. High Missingness Columns Dropped

3 columns exceeding 40% missingness (weight, medical_specialty, payer_code) were removed.

3. Transformation of Skewed Features

Extreme skewness in utilization features required log1p transformations for normalization.

4. Near-Zero Variance Reduction

16 medication columns exhibiting near-zero variance (>97% uniform) were dropped to reduce noise.

5. Strongest Predictor Signal

Prior hospitalization history emerged as the most prominent indicator of readmission risk.

6. High Cardinality Addressed

Over 700 unique ICD-9 diagnosis codes were successfully grouped into 10 broader clinical categories.

7. Meaningful Outlier Preservation

Extreme values represent genuine high-risk patients; they were winsorized rather than removed.

Key Takeaway

A clear, robust data strategy has been established, setting a solid foundation for the modeling phase.

Preprocessing Step 1: Data Cleaning

1. **Drop Identifiers:** encounter_id, patient_nbr *(No predictive value)*
2. **Drop >40% Missing:** weight, payer_code, medical_specialty *(Noise)*
3. **Drop Near-Zero Variance:** 16 medication columns *(Near-zero entropy)*
4. **Filter Rows:** Remove deceased/hospice (1,545), invalid gender (3) *(Validity)*
5. **Impute Remaining:** race → mode, diagnoses → 'Missing' *(Completeness)*

Shape Tracking

Rows

71,518 → 69,970

Columns

50 → 29

Key Takeaway

Rigorous data cleaning significantly reduced feature space dimensionality and improved dataset quality.

Preprocessing Step 2: Feature Encoding

Feature Type	Encoding Method	Example
Ordinal (age)	Numeric midpoints	[70-80) → 75
Ordinal (meds)	Integer mapping	No=0, Steady=1, Down=2, Up=3
Ordinal (labs)	Integer mapping	None=0, Norm=1, >7=2, >8=3
Binary	0/1 mapping	Male=1, Female=0
Nominal (race)	One-hot encoding	race_Caucasian, race_Other...
High-card (ICD9)	Group → one-hot	250.xx → Diabetes → diag_1_Diabetes

Dimensionality Shift

Feature Columns

29 → 121

Expansion primarily driven by one-hot encoding of categorical variables.

Key Takeaway

All features are now fully numeric and mathematically scaled, enabling distance-based and tree-based ML algorithms.

Reducing Diagnosis Code Cardinality

Before: 700+ Unique Codes

250.01, 250.02, 250.13...
(All specific diabetes variations)

410.0, 414.1, 428.0...
(All circulatory conditions)

Creates extreme data sparsity

Leads to the "Curse of Dimensionality"

Reduces model generalization

After: 10 Clinical Categories

Circulatory(390-459, 785)
Respiratory(460-519, 786)
Digestive(520-579, 787)
Diabetes(250.xx)
Injury(800-999)
Musculoskeletal(710-739)
Genitourinary(580-629, 788)
Neoplasms(140-239)
Other
Missing

Key Takeaway

Grouping high-cardinality diagnosis codes reduces sparsity and helps models learn generalizable clinical patterns.

Domain-Driven Feature Construction

New Feature	Formula / Construction	Clinical Rationale
total_visits_prior	outpatient + emergency + inpatient	Captures aggregate utilization intensity; identifies "frequent flyers" across all clinical settings.
n_med_changes	count(meds == 'Up' | meds == 'Down')	Indicates treatment instability or disease progression during the current hospital stay.
a1c_tested	1 if A1Cresult != 'None' else 0	Binary indicator of whether clinical attention was given to long-term glycemic control.
glu_tested	1 if max_glu_serum != 'None' else 0	Binary indicator of acute monitoring intensity for blood glucose during admission.
service_utilization	num_lab_procedures + num_procedures	Total volume of interventions during the encounter; proxy for case complexity.
stay_length	Binned: Short(1-3d), Med(4-7d), Long(8+)	Categorizes admission duration to capture non-linear risk associated with extended stays.

Key Takeaway

Combining simple variables into domain-specific features often provides stronger signals than raw data alone.

Dimensionality Management

1. Log Transform

Applied to skewed features to normalize distributions for distance-based models.

2. Variance Threshold Filtering

Removed 29 near-zero variance one-hot columns (var < 0.01) to eliminate low-signal features.

3. Multicollinearity Pruning

Dropped 4 redundant features (|r| > 0.95), such as number_inpatient vs. number_inpatient_log.

Final Shape

Features Reduced

132 → 98

Achieving a balanced feature count for robust and efficient modeling.

Key Takeaway

A 3-step dimensionality reduction process successfully condensed the feature space, improving model efficiency.

Stratified 80/20 Train-Test Split

Dataset Partitioning (~70k total)

Full Dataset (100%) - ~9% Positive

Training Set (80%)
~56,000 rows | ~9% Positive

Test (20%)
~14k | 9%

Stratification is Critical

With only ~9% positive cases, random splitting risks creating subsets with disproportionate class representation. Stratified splitting ensures both sets maintain the true real-world distribution.

Data Leakage Safeguards

- Patient Deduplication (Section 4)
- Stratified split preserves class ratio
- Scaler fit on training data ONLY

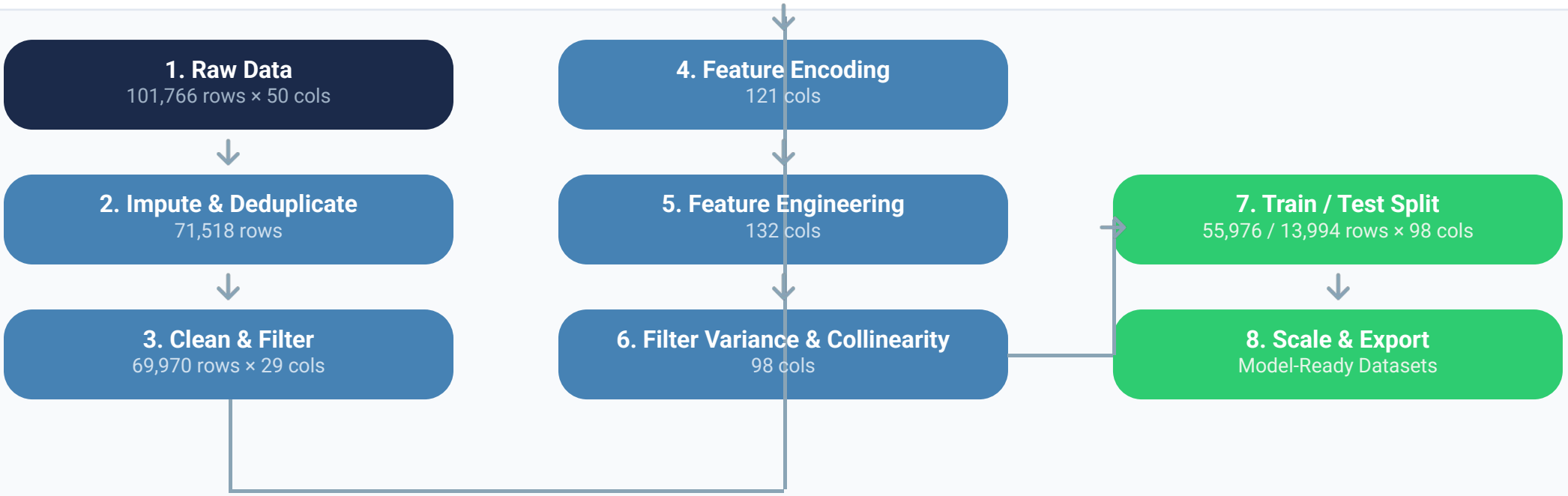
Phase 1 Outputs Saved

Both **scaled** (for SVM, Logistic Regression, Neural Nets) and **unscaled** (for Decision Trees, Random Forests) versions of train/test sets successfully exported.

Key Takeaway

Rigorous splitting methodology ensures out-of-sample evaluation in Phase 2 will be statistically valid and free from data leakage.

From Raw Data to Model-Ready Features



Key Takeaway

A systematic, documented, and fully reproducible pipeline ensures no data leakage and guarantees consistent preprocessing for all new data.

Phase 1 Conclusions

Data Transformation

Successfully transformed raw clinical data (101K × 50) into a clean, model-ready dataset (70K × 98 features).

Leakage Prevention

Applied rigorous data leakage safeguards including patient deduplication, train-only scaling, and stratified splitting.

Predictive Signals

Identified prior hospitalization and utilization history as the strongest univariate predictor family for readmission.

Feature Engineering

Built 5 domain-informed engineered features to capture complex clinical patterns and treatment intensity.

Dimensionality Management

Controlled feature explosion: expanded from 50 raw to 121 encoded, then filtered down to 98 robust features.

Evaluation Strategy

Documented severe class imbalance (~9% positive). Established Precision, Recall, and F1 as primary metrics.

Key Takeaway

Phase 1 has established a solid, reliable, and high-quality data foundation ready for Phase 2 predictive modeling.

Coming in Phase 2: Modeling & Evaluation

Planned Models

- **Decision Tree** (no scaling needed)
- **Logistic Regression** (scaled features)
- **Naive Bayes**
- **Support Vector Machine** (scaled features)

Evaluation Metrics

- **Precision & Recall** (key for imbalance)
- **F1-Score** (primary harmonic mean metric)
- **AUC-ROC** (evaluates varied thresholds)
- **Calibration Assessment**

Validation Process

- **Cross-validation** for reliable model selection
- **Hyperparameter tuning** (Grid/Random search)
- **Threshold optimization** for recall enhancement

Optional Extensions

- **K-Means clustering**
- **Silhouette score evaluation**
- **Cohort discovery** to understand patient clusters

Key Takeaway

The next step is systematic model selection and tuning, leveraging the clean, leakage-free dataset prepared in Phase 1.

References

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Key Takeaway

Our methodology and analytical framework are solidly grounded in peer-reviewed literature and established statistical learning theory.